Product *B*

Sales(x)	Frequency(f)	A= 225 $d = x - A$	fd^2	fd^2
125	1	-100	-100	10,000
150	1	-75	-75	5625
200	1	-25	-25	625
225	1	0	0	0
250	1	25	25	625
310	1	85	85	7225
330	1	105	105	11,025
Total	**7**		**15**	**35,125**

$$\bar{x} = A + \frac{\sum fd}{N}$$

$$= 225 + \frac{15}{7}$$

$$= 225 + 2.14 = 227.14$$

$$\text{Variance } \sigma^2 = \frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2$$

$$= \frac{35125}{7} - \left(\frac{15}{7}\right)^2$$

$$= 5017.85 - 4.59$$

$$= 5013.26$$

$$\text{Standard deviation} = \sqrt{5013.26} = 70.80$$

$$\text{Coefficient of variation } (C.V)\ B = \frac{70.80}{227.14} \times 100$$

$$= 31.17\%$$

Since the coefficient of variation for product A is more than that of product B,

Therefore the fluctuation in sales of product A is higher than product B.

6.6 Moments

6.6.1 Raw moments:

Raw moments can be defined as the arithmetic mean of various powers of deviations taken from origin. The rth Raw moment is denoted by μ_r', r = 1,2,3... Then the first raw moments are given by

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